

KUNJ VAGHANI

B.Tech. Computer Science and Engineering
Sardar Vallabhbhai National Institute of Technology, Surat

+91-7990806810
kunjvaghani864@gmail.com
[in](#) [kunj-vaghani](#) [kunjvaghani](#)
[Portfolio](#)

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Sardar Vallabhbhai National Institute of Technology, Surat	CGPA: 7.83	2027
Higher Secondary	P.P. Savani Vidhyalaya, Surat (GSEB)	79.00%	2023

SKILLS

- Languages:** C++, Python, JavaScript, SQL
- Web Technologies:** HTML, CSS, Express.js, Node.js, React.js, Next.js, Tailwind CSS, RESTful APIs, Streamlit
- Databases:** MySQL, MongoDB, Supabase, ChromaDB
- AI/ML:** LangChain, Retrieval-Augmented Generation (RAG), Hugging Face, OpenAI Embeddings, Re-Ranking
- Core CS Fundamentals:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Prompt Engineering
- Tools:** Git, GitHub, Postman, n8n

PROJECTS

- RAG Intelligence: Document Intelligence Platform** [Demo](#)
Python, LangChain, ChromaDB, HuggingFace Transformers, Mistral AI, OpenAI API, Streamlit, PyMuPDF
 - Architected an end-to-end Retrieval-Augmented Generation (RAG) pipeline supporting multi-format document ingestion (PDF, DOCX, HTML, Markdown) using PyMuPDF and LangChain's RecursiveCharacterTextSplitter.
 - Engineered a hybrid retrieval system combining Maximal Marginal Relevance (MMR) with Cross-Encoder re-ranking (ms-marco-MiniLM-L-6-v2), filtering up to 66% of low-relevance chunks below a 0.1 threshold to maximize LLM context quality.
 - Integrated multi-model LLM backend (Mistral AI, Google Gemini) with real-time token-by-token streaming via Streamlit and a context transparency viewer displaying source citations, cross-encoder relevance scores, and page references, enabling interpretable and hallucination-resistant AI responses.
- SecureVerify AI: Synthetic Identity Fraud Detection** [Demo](#)
Python, Streamlit, OpenCV, DeepFace, TensorFlow, Tesseract OCR, MongoDB
 - Developed an AI-driven e-KYC system to detect synthetic identity fraud by correlating document authenticity, facial biometrics, and behavioral signals using multi-modal data fusion.
 - Implemented document forgery detection using Error Level Analysis (ELA) and EfficientNet, achieving good accuracy in identifying AI-generated and tampered government IDs.
 - Engineered a biometric verification module with DeepFace embeddings and Laplacian-based liveness detection to prevent deepfake and spoofing attacks.
- MockMate: AI-Powered Interview Simulator** [Demo](#)
TypeScript, Next.js, React, Prisma, MongoDB, Google Gemini API, Clerk, Recharts
 - Engineered an AI-powered interview simulation platform using Google Gemini 1.5 API to generate dynamic, multi-turn context-aware questions tailored to target job roles and experience levels.
 - Integrated browser-native Web Speech API for real-time speech-to-text and text-to-speech interaction, eliminating dependency on paid cloud voice services and reducing operational costs.
 - Developed an advanced analytics dashboard with Recharts radar charts to quantitatively evaluate candidates across 5 dimensions - Technical, Communication, Problem Solving, Experience, and Confidence with per-question AI-generated feedback.

POSITIONS OF RESPONSIBILITY

- Co-Head, Technical Committee, MINDBEND** *Event Organizer / Nov 2024 - April 2025*
 - Designed and led CodeCraft at MINDBEND 2024, managing participant shortlisting, final-round execution, and prize distribution under tight timelines.
- Student Coordinator, BIS Student Chapter, CSE Department** *Event Organizer / Sep 2025 - present*

ACHIEVEMENTS & CERTIFICATIONS

- ACM Summer Challenge:** Completed a 30-day advanced DSA coding challenge, implementing optimized algorithmic solutions. Earned a [Certificate](#).
- Echelon Hackathon:** Participated in a competitive hackathon and built an AI-driven e-KYC system for Synthetic Identity Fraud Detection using multi-modal data fusion (Document, Biometric, Behavioral). Earned a [Certificate](#).

PROFILES AND LINKS

- LeetCode:** Solved **650+** Problems, Rating: **1524** | [kunjagghani](#)
- Codeforces:** Solved **290+** Problems, Max Rating: **1280** | [VK66](#)
- CodeChef:** 3 Star Coder, Max rating: **1618** | [kunj_vaghani](#)
- GeeksforGeeks:** Solved **180+** Problems | [kunjagh5z1t](#)